SHIKSHA SETU

LEARNING MANAGEMENT SYSTEM

Technical & Business Proposal

Submitted to the Department of Education

Version 1.0 | 2025 | Confidential

7 Core Modules	500K+ Users	99.5% SLA	NEP 2020 Ready

DOCUMENT CONTROL

Version	Date	Prepared By	Status
1.0	2025	SS	Final Draft

1. SUMMARY

Shiksha Setu's Learning Management System (LMS) is a comprehensive, enterprise-grade, cloud-native digital education platform purpose-built for large-scale deployment across schools, colleges, and government institutions. It centralizes the full lifecycle of teaching and learning — from course creation and content delivery to assessments, analytics, and stakeholder communication — into a single, unified, secure, and scalable ecosystem.

The platform is designed from the ground up to address the unique operational, regulatory, and pedagogical requirements of state-run educational institutions, enabling standardization across hundreds of institutions while preserving the flexibility required by individual schools and colleges to manage their own academic operations.

This document serves as a combined Technical and Business Proposal, providing the Department of Education with all necessary information to evaluate, procure, and deploy the Shiksha Setu LMS at state-wide scale. It covers platform architecture, security posture, integration capabilities, implementation roadmap, support structure, and commercial framework.

Key Highlights at a Glance

Parameter	Details
Platform Type	Cloud-Native, Multi-Tenant LMS
Deployment Model	SaaS / Private Cloud / Hybrid
Target Users	Students, Teachers, Admins, District/State Officials
Functional Modules	7 Core Modules
Authentication	OTP-based (Devise) + JWT
Encryption Standard	AES-256 (at rest) + TLS 1.2+ (in transit)
Uptime SLA	99.5%+ Monthly
Compliance	IT Act 2000, CERT-In, NEP 2020, PDPA-aligned
Integration	REST API / GraphQL / SSO / Bulk Data Import
Scalability	Up to 500,000+ concurrent users (scaled)

2. PLATFORM OVERVIEW & VISION

2.1 What is the Shiksha Setu LMS?

The Shiksha Setu Learning Management System is an integrated digital education platform engineered to transform how students, teachers, and administrators interact with educational content. Whether deployed in a rural middle school or a large state university, the platform provides consistent, curriculum-aligned, and data-driven learning experiences across the board.

The LMS is built on the principle that effective education technology must be invisible in its operation — it should reduce administrative friction, surface actionable insights, and create more time for meaningful teaching and learning, not add complexity.

2.2 Core Objectives

Consistency at Scale: Every school and college on the platform follows the same standardized workflows for course creation, assessment, reporting, and compliance — ensuring uniform quality regardless of institution size or location.

Personalized Learning Delivery: Content dripping, adaptive pathways, and learner tracking allow each student to progress at the appropriate pace while giving teachers early visibility into at-risk learners.

Administrative Efficiency: Role-based dashboards and automated processes for attendance, grading, and notifications eliminate manual overhead, allowing teachers and administrators to focus on their core responsibilities.

Data-Driven Governance: Real-time analytics accessible to District Education Officers and state administrators enable evidence-based policy decisions, targeted resource allocation, and proactive institutional interventions.

Seamless Integration: The platform is designed to work alongside — not replace — existing state MIS, ERP, and payment systems, connecting through standard APIs to minimize duplication of effort.

2.3 Platform Philosophy

The LMS operates on a multi-tenant architecture where each institution is logically isolated within a shared cloud infrastructure. This means state administrators get consolidated visibility across all institutions, while each school or college operates in its own secure, independently configurable environment. No institution's data is accessible to another.

3. TECHNICAL SPECIFICATIONS & ARCHITECTURE

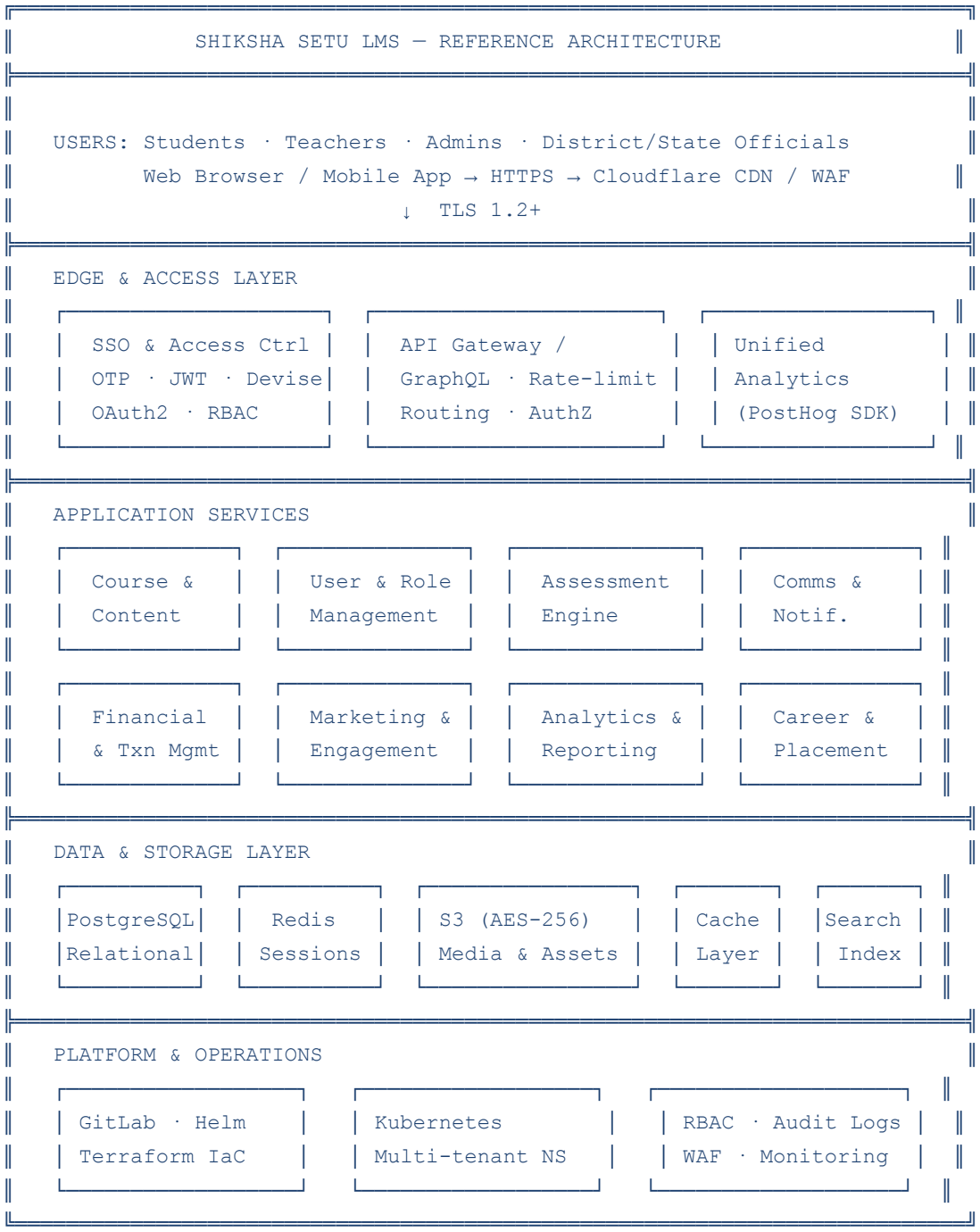
3.1 Technology Stack

The Shiksha Setu LMS is built on a mature, production-proven technology stack selected for reliability, security, developer ecosystem strength, and long-term maintainability.

Layer	Technology	Role
Marketing / Info Pages	GatsbyJS (React SSG)	Static marketing and public-facing pages
LMS Frontend	Next.js (React Framework)	Student, teacher, and admin UI
Admin Dashboard	Ruby on Rails + Web Technologies	Platform administration and configuration
Backend API	Ruby on Rails	Business logic, REST + GraphQL API endpoints
Primary Database	PostgreSQL	Relational data — users, courses, enrollments, grades
Document / Session Store	Redis	Session management, caching, rate limiting
Search & Indexing	MeiliSearch	Fast, typo-tolerant content and course search
Object Storage	Amazon S3 (AES-256 encrypted)	Course assets, media files, certificates
CDN / Edge Layer	Cloudflare	Global content delivery, DDoS protection, WAF
Email Notifications	Postmark	Transactional email delivery
Payment Gateway	Razorpay + Chargebee	Fee collection and subscription billing
CI/CD Pipeline	GitLab Pipelines + Helm	Automated build, test, and deployment
Container Orchestration	Kubernetes (multi-tenant namespaces)	Service deployment, autoscaling, isolation
Infrastructure as Code	Terraform	Reproducible, version-controlled infrastructure
Analytics SDK	PostHog	Usage analytics, feature tracking, funnel analysis

3.2 High-Level System Architecture

The architecture follows a layered design pattern with clear separation between the user access layer, application services, data storage, and platform operations.



3.3 Multi-Tenant Architecture

The LMS employs a logical multi-tenancy model where each institution (school, college, or district) operates in an isolated namespace within the shared Kubernetes cluster.

Data Isolation: No cross-tenant data access is architecturally possible. Each institution's PostgreSQL schema, S3 bucket prefixes, and application-level access controls are independently governed.

Independent Configuration: Each institution can maintain its own branding, academic calendar, curriculum structure, grading rubrics, and user roles without affecting any other institution on the platform.

Consolidated Oversight: State and district administrators are provisioned with cross-tenant read access, enabling analytics and reporting across all institutions under their jurisdiction — without direct data access to individual institution records.

Resource Governance: Kubernetes namespace-level resource quotas ensure that a spike in usage from one institution does not degrade performance for others on the same infrastructure.

3.4 API Architecture

All platform functionality is exposed through a versioned API layer, enabling third-party integrations, state MIS connectivity, and programmatic access for authorized systems.

- Protocol: REST (primary) + GraphQL (for dashboard data queries)
- Authentication: JWT Bearer tokens issued post OTP verification
- Rate Limiting: Per-IP and per-token rate limiting enforced at the API gateway layer
- Versioning: URI-based versioning (e.g., /api/v1/, /api/v2/) with backward compatibility guaranteed across minor versions
- Documentation: OpenAPI 3.0 (Swagger) documentation available for all endpoints
- Webhook Support: Real-time event notifications (enrollment, submission, completion) via configurable webhooks to connected state systems

4. FUNCTIONAL MODULES — DETAILED BREAKDOWN

The Shiksha Setu LMS is organized into seven major functional modules. Each module is independently configurable while sharing a common authentication, analytics, and notification infrastructure.

4.1 Course Management

This is the core instructional delivery module. It handles the full lifecycle of learning content — from creation and scheduling to delivery and archival.

Courses: Instructors can create standalone course units with customizable metadata, prerequisite structures, target audience settings, and enrollment controls. Each course supports multiple content formats including video, PDF, audio, slide decks, interactive HTML, and external embeds.

Smart Content Dripping: Course administrators can configure time-based or milestone-based content release schedules, ensuring that curricula unfold in logical, pedagogically sound sequences aligned with the academic calendar. This is particularly valuable for semester-structured college programs and grade-wise school syllabi.

Series & Learning Paths: Related courses can be grouped into structured series or learning pathways (e.g., "Class 10 Science Series" or "First-Semester Engineering Foundation"). Series support prerequisite enforcement, meaning students must complete earlier modules before progressing.

Live Sessions: The platform natively supports scheduling, hosting, and recording of live virtual classes and webinars. Live session links are auto-distributed to enrolled students via in-app notifications and email. Post-session recordings are automatically archived within the course module.

User Course Tracking: Granular per-student enrollment and progress data, including content completion percentages, time-on-task metrics, and last-accessed timestamps, are maintained for every course-student combination.

4.2 User & Role Management

The platform supports a hierarchical, multi-level role structure designed to mirror the governance model of a state education system.

Role Level	Scope	Key Permissions
State Administrator	All institutions in the state	Platform-wide settings, curriculum frameworks, cross-institution analytics
District Administrator	All institutions in a district	District-level oversight, institution onboarding, aggregated reporting
Institution Admin (Principal / Coordinator)	Single school or college	User management, course assignments, institution-level configuration
Instructor / Teacher	Assigned courses	Content creation, grading, live session management, class analytics
Student / Learner	Enrolled courses	Content access, assignment submission, live participation, progress tracking

All role assignments are managed through an administrative interface with full audit logging. Role permissions are additive and non-inheritable — a District Administrator role does not implicitly grant Institution Admin rights at a specific school.

4.3 Financial & Transaction Management

For institutions operating fee-based programs, vocational certifications, or adult education courses, the platform provides a built-in financial management module.

Transactions: All fee payments — whether for regular tuition, specific course enrollments, or certification programs — are recorded within the platform, linked to the corresponding user and course, and available for export and audit.

Subscription Plans: Flexible plan structures can be configured for semester-based recurring billing, annual subscriptions, or one-time course purchases. Razorpay and Chargebee handle the payment gateway and subscription lifecycle respectively.

Coupon & Discount Management: Promotion codes with configurable discount amounts, usage limits, and validity periods can be created by administrators — useful for scholarship programs, merit discounts, or government-sponsored access drives.

Financial Reporting: Exportable transaction ledgers in Excel and PDF formats are available to institution finance personnel and state-level auditors.

4.4 Marketing & Engagement

Lead Management: The platform can capture and manage inquiries from prospective students for government-run diploma programs, distance learning initiatives, or skill certification courses.

Affiliate Tracking: Referral sources — including career counselors, community outreach programs, and employment cells — can be registered as affiliates, with enrollment attribution tracked per referral source.

Community Forums: Each course or institution can have associated discussion forums and peer groups. Community spaces are moderated by instructors and support threaded discussions, file sharing, and pinned announcements.

4.5 Learning Pathways & Career Support

Skill Courses Catalog: A searchable catalog of state-approved upskilling courses, remedial programs, and vocational offerings is maintained separately from the core curriculum, allowing students to self-enroll in supplementary content.

Internship Program Management: The platform provides a structured module to list, apply for, and manage internship and field-visit opportunities, connecting college departments with industry partners and students with hands-on learning experiences.

Job Portal Integration: Graduating students can be directed to integrated state employment portals or directly connected job vacancy listings. The LMS can surface relevant opportunities based on completed courses and certified skills.

4.6 Communication & Notifications

Multi-Channel Notifications: The platform supports SMS, email, and in-app push notifications, configurable per notification type and user preference. All notifications are templated and support variable substitution for personalization.

Automated Alerts: Configurable triggers for events including assignment deadlines, exam timetables, attendance thresholds, scholarship application windows, and session reminders — all dispatched without manual intervention.

Broadcast Messaging: State and district administrators can issue platform-wide or institution-specific broadcast messages to all users within their scope.

4.7 Analytics & Reporting

Real-Time Dashboards: Each role level has access to analytics dashboards appropriate to their scope — from individual teacher class performance summaries to state-level enrollment and completion heatmaps.

Learner Risk Identification: Built-in algorithms flag students with declining engagement patterns (drop in content access, missed assignments, low quiz scores) to enable early intervention by teachers and counselors.

Standard Reports Available:

- Attendance summary (daily, weekly, monthly)
- Course completion rates by institution / district / state
- Assignment submission and grading status
- Assessment performance distribution
- Fee collection and transaction summary
- Live session participation logs
- Content engagement heatmaps

Export Formats: All reports are exportable in Excel (.xlsx) and PDF formats for offline review, audit submission, and external BI tool ingestion.

5. ROLE-SPECIFIC DASHBOARDS

Each user type is served a purpose-built dashboard that surfaces only the information and controls relevant to their responsibilities. This role-based view architecture minimizes cognitive load and reduces the risk of unauthorized data exposure.

5.1 State / District Administrator Dashboard

The state-level dashboard provides a consolidated, bird's-eye view of the entire education network under the administrator's jurisdiction.

Analytics Panel: Aggregated daily active user counts, course completion rates, assessment pass percentages, and attendance trends across all institutions — filterable by district, institution type, subject, or time period.

Institution Management: View and manage all onboarded institutions, their configuration status, active user counts, and compliance metrics.

Curriculum Governance: Define and enforce curriculum frameworks, academic calendars, grading rubrics, and content standards that apply across all institutions.

Financial Overview: Consolidated fee collection data across all fee-based programs within the state or district.

Alert & Escalation Feed: Automated flags for institutions with below-threshold completion rates, high absenteeism, or low teacher engagement — enabling targeted intervention.

5.2 Institution Administrator Dashboard (Principal / Coordinator)

The institution-level dashboard gives principals and academic coordinators complete operational control over their school or college within the platform.

Teacher Management: View all registered teaching staff, their assigned courses, content upload status, and live session schedules.

Student Enrollment Console: Manage bulk enrollment, individual student access, batch transfers, and course-level capacity settings.

Performance Overview: Institution-level performance analytics including average assessment scores by class or department, course completion rates, and engagement metrics.

Notification Center: Send targeted announcements to students, parents (where configured), or teachers within the institution.

5.3 Instructor / Teacher Dashboard

The teacher dashboard is designed for day-to-day instructional workflow, minimizing navigation and surfacing the most immediate action items.

My Courses Feed: A prioritized view of all active courses assigned to the instructor, with quick-access links to content management, student roster, and grading queue.

Assignment & Assessment Queue: Pending submissions requiring grading are surfaced automatically, with inline grading tools for subjective and objective assessments.

Live Session Management: Create, schedule, and launch live sessions directly from the dashboard, with pre-session participant visibility.

Class Analytics: Per-course performance data including average quiz scores, assignment submission rates, content engagement heatmaps, and at-risk student flags.

5.4 Student / Learner Dashboard

The student dashboard is optimized for ease of access across devices, including low-spec tablets and mobile phones — critical for equitable access in government school contexts.

My Courses: A clean, visual listing of all enrolled courses with progress indicators, upcoming deadline callouts, and continue-where-you-left-off navigation.

Assignment Tracker: Consolidated view of all pending, submitted, and graded assignments across all enrolled courses.

Live Session Access: Upcoming live class schedule with single-click join links.

Community & Discussions: Access to course-specific forums, peer groups, and instructor announcements.

Certificate Gallery: All earned completion certificates, downloadable in PDF format with verifiable unique identifiers.

6. LICENSING MODEL

6.1 Licensing Options

The Shiksha Setu LMS is available under multiple licensing structures to align with state procurement preferences, budget cycles, and deployment scale.

Model	Description	Best For	Billing Cycle
Per Student License	Annual per-learner fee based on active enrolled student count. All features included.	Large state deployments, enrollment-linked budgeting	Annual
Per Institution License	Fixed annual fee per institution (school / college) regardless of student headcount.	Districts with fixed-size institutions	Annual
Enterprise / State License	Flat fee covering all institutions within a defined state or district, up to an agreed cap.	State-wide rollouts, maximum cost predictability	Annual or 3-Year
One-Time + AMC	Perpetual license with Annual Maintenance Contract for updates, patches, and support.	Institutions preferring CAPEX over OPEX expenditure	One-Time + Annual AMC

6.2 What Is Included in Every License Tier

Irrespective of the licensing model selected, the following are included as standard:

- Full access to all seven LMS functional modules and role-specific dashboards
- Unlimited course creation and content uploads per licensed institution
- Standard third-party integrations: Postmark (email), Razorpay (payments), Cloudflare CDN
- Software updates, minor version releases, and security patches throughout the active license period
- 99.5%+ monthly uptime SLA with tiered support access
- Multi-tenant data isolation and RBAC governance
- All reports and analytics capabilities

6.3 Optional Add-On Modules (Separately Licensed)

The following modules from the broader Shiksha Setu ecosystem can be integrated with the LMS under separate licensing:

- 3D Scholar — AR/VR immersive STEM content
- AI Teacher Evaluation — Automated handwritten and digital answer sheet grading
- Virtual Labs — Cloud-based science experiment simulations
- ERP and Library for Schools — Fee, payroll, attendance, and administrative governance

7. IMPLEMENTATION TIMELINE & PROJECT PLAN

7.1 Phased Rollout Approach

The deployment of the Shiksha Setu LMS follows a structured five-phase implementation methodology. This phased approach ensures controlled risk, early feedback incorporation, and progressive scaling from pilot institutions to full state-wide deployment.

Phase 1 — Discovery & Requirements (Weeks 1–2)

Conduct structured stakeholder workshops with Department of Education representatives, district IT teams, and representative teachers to document:

- Institutional hierarchy and reporting structure
- Existing MIS/ERP systems requiring integration
- Curriculum frameworks and academic calendar alignment
- User volume projections (students, teachers, admins) per district
- Data formats for bulk user and institution import

Deliverable: Signed Requirements Specification Document and Architecture Decision Record.

Phase 2 — Environment Setup & Configuration (Weeks 3–5)

- Provision cloud infrastructure (Dev, UAT, Production environments) via Terraform
- Configure Kubernetes namespaces per institutional tier
- Set up domain and subdomain structure with SSL certificates
- Apply institutional branding (logos, color palette, custom domain if required)
- Configure SSO integration with state education portal (if applicable)
- Establish CI/CD pipeline with environment-gated deployment approvals

Deliverable: Fully provisioned Dev and UAT environments; infrastructure documentation.

Phase 3 — Pilot Launch — Seed Institutions (Weeks 6–10)

- Onboard 5 to 10 representative pilot institutions (mix of urban and rural, school and college)
- Conduct administrator and teacher training at pilot institutions
- Facilitate initial content upload and course structuring by pilot teachers
- Run live sessions and assessments with real student cohorts
- Collect structured feedback via in-platform surveys and direct interviews
- Incorporate critical feedback into configuration; document enhancements for Phase 4

Deliverable: Pilot Go-Live Report; feedback incorporation log; updated training materials.

Phase 4 — State-Wide Rollout (Weeks 11–20)

- District-by-district onboarding in agreed sequence with DoE
- Bulk institution and user creation via import tools
- Parallel training batches for administrators and teachers at scale (Train-the-Trainer model)
- Live session infrastructure load testing at production scale
- State and district administrator dashboard activation
- Communication to all students and guardians via institution channels

Deliverable: Full state-wide go-live; per-district onboarding completion report.

Phase 5 — Stabilization & Handover (Weeks 21–24)

- 24×7 heightened monitoring during initial post-launch period
- Rapid bug triage and resolution
- Knowledge transfer sessions for state IT team
- Documentation handover: operations runbook, admin guide, API reference
- SLA period officially commences

Deliverable: SLA activation letter; knowledge transfer completion certificate; project closure report.

7.2 Key Milestones

Milestone	Target Week	Owner	Sign-off
Contract Signing & Project Kickoff	Week 1	Both Parties	Yes
Requirements Document Signed	Week 2	Both Parties	Yes
Infrastructure Live (Dev + UAT)	Week 3	Shiksha Setu	Yes
Production Environment Ready	Week 5	Shiksha Setu	Yes
Pilot Institutions Go-Live	Week 10	Shiksha Setu + DoE	Yes
Full State-Wide Go-Live	Week 20	Both Parties	Yes
Stabilization Complete	Week 24	Shiksha Setu	Yes
SLA Period Commences	Week 21	Shiksha Setu	Yes

7.3 Prerequisites & Dependencies

The following inputs are required from the Department of Education for the timeline above to hold:

- Institution master list (name, type, district, address) in agreed format
- Student and teacher roster data in CSV / Excel format
- Existing MIS or ERP API documentation (if SSO or data sync is required)
- Approval for cloud deployment region or state data center access (for private cloud option)
- Nominated state-level Project Coordinator and District Nodal Officers

8. DEPLOYMENT & INTEGRATION APPROACH

8.1 Deployment Options

Option	Description	Recommended For
SaaS — Cloud Hosted	Fully managed by Shiksha Setu on AWS-compatible infrastructure. State accesses via branded subdomain. Zero infrastructure overhead for the state.	Most state deployments; fastest time-to-launch
Private Cloud — State Data Center	Software deployed within a state-approved data center or NIC/STPI facility. Shiksha Setu manages the software layer remotely.	States with data residency or sovereignty requirements
Hybrid	Core application hosted on state infrastructure; media and CDN assets served via Cloudflare edge nodes for performance optimization.	Large states with mixed connectivity; partial data sovereignty needs

8.2 Environment Architecture

Three separate environments are maintained throughout the lifecycle of the platform:

Development (Dev): Used exclusively by Shiksha Setu engineering. All feature development and unit testing occurs here before promotion. No state data present in this environment.

UAT (User Acceptance Testing): A production-equivalent environment used for state IT team validation, teacher acceptance testing, and integration testing with connected state systems. Populated with anonymized or synthetic data.

Production: The live environment accessed by all end users. Protected by strict change management processes — all changes require successful UAT sign-off and go through the GitLab CI/CD deployment pipeline with manual approval gates before reaching production.

8.3 Branding & Domain Configuration

Each deployment is fully whitelabeled with institutional branding. Standard subdomain structure:

- lms.[domain] — Primary student and teacher portal
- admin.[domain] — Administrative dashboard
- api.[domain] — Secure API endpoint for integrations

Custom government domain support is available (e.g., lms.telanganaedu.gov.in) subject to DNS delegation by the state. All subdomains are provisioned with auto-renewing SSL/TLS certificates via Cloudflare, ensuring HTTPS enforcement at all times.

8.4 System Integration Capabilities

Integration	Protocol	Purpose
State MIS / SIS	REST API / CSV bulk import	Student and teacher roster synchronization
SSO / Identity Provider	OTP + JWT / OAuth 2.0 / SAML 2.0	Single sign-on from state education portal
Payment Gateway	Razorpay + Chargebee API	Fee collection and subscription lifecycle
Email Service	Postmark SMTP / API	Transactional and bulk notification delivery
Video Hosting	Vimeo API / S3 + CloudFront	High-quality lecture streaming and recording
State Employment Portal	REST API	Job and internship program integration
Analytics / BI Tools	GraphQL / CSV export / Webhooks	Data feeds to state BI and reporting dashboards
ERP / School Admin System	REST API	Fee and attendance data exchange

8.5 Data Migration

For institutions with existing student records, course content, or historical assessment data, Shiksha Setu provides a structured data migration service:

- Standardized data import templates provided for users, courses, and enrollment records
- Bulk import tools with validation, error reporting, and rollback capability
- Content migration support for existing video, PDF, and document libraries
- Historical grade and attendance data import (subject to source format compatibility)

9. TRAINING & CAPACITY BUILDING

9.1 Training Philosophy

Shiksha Setu follows a "Train the Trainer" model as the primary delivery mechanism for large-scale state deployments, combined with direct end-user onboarding workshops at pilot institutions and self-paced digital training modules embedded within the LMS itself.

This approach is designed to create a sustainable, self-sufficient training ecosystem within the state — where trained master trainers (district education officers, senior teachers, or IT coordinators) can continue onboarding new users without repeated vendor dependency.

9.2 Training Programs

Program	Target Audience	Duration	Mode
Platform Administrator Training	State / District IT Admins	2 Full Days	In-Person or Virtual
Institution Admin Training	Principals, Academic Coordinators	1 Day	In-Person or Virtual
Teacher / Instructor Training	All Teaching Staff	4 Hours	Group Workshop / Self-Paced Module
Student Orientation	All Enrolled Students	1 Hour	Self-Paced Module (within LMS)
Content Creator Workshop	Subject Matter Experts, Senior Teachers	1 Day	Hands-On Workshop
Analytics & Reporting Training	District Officers, DoE Officials	3 Hours	Virtual Session

9.3 Training Deliverables

The following materials are provided as part of the training engagement:

Documentation Package:

- Role-specific user manuals in PDF format (available in English and regional language)
- Quick reference cards (laminated print-ready format) for teachers and administrators
- Troubleshooting guide for common issues and self-service resolution

Digital Training Assets (hosted within the LMS):

- Recorded video tutorial library organized by role and feature area
- Interactive walkthrough modules with progress tracking
- Assessment module to verify training completion (with certificates issued by the LMS)

Post-Training Support:

- Dedicated helpdesk support line during the first 90 days post-launch
- Quarterly refresher webinars for new features and platform updates

- Community forum for master trainers to share learnings and resolve peer queries

9.4 Language & Accessibility

Training materials and the LMS interface are configurable to support regional languages including Telugu, Hindi, and English. The platform's UI supports Unicode rendering for all official Indian languages, ensuring accessibility for teachers and students across all linguistic demographics in the state.

10. SUPPORT & MAINTENANCE (AMC / CMC)

10.1 Support Structure

Shiksha Setu operates a four-tier support model designed to match response intensity to issue criticality, ensuring that minor queries do not consume capacity reserved for platform-level emergencies.

Tier	Channel	Coverage	Response Time
Tier 1 — User Helpdesk	Web Portal + Email	Monday–Saturday, 8 AM – 8 PM	< 4 Business Hours
Tier 2 — Technical Support	Email + Phone + Remote Desktop	Monday–Saturday, 9 AM – 6 PM	< 8 Business Hours
Tier 3 — Critical Escalation	Direct Engineering Team	24 × 7 (Critical Issues Only)	< 2 Hours
State Nodal Officer Hotline	Dedicated Phone + WhatsApp	Monday–Saturday, 9 AM – 6 PM	Immediate

10.2 AMC / CMC Scope

The Annual Maintenance Contract (AMC) — or Comprehensive Maintenance Contract (CMC) for deployments with infrastructure coverage — includes the following as standard:

Software Maintenance:

- All minor version updates, security patches, and bug fixes at no additional cost during the active AMC period
- Quarterly feature releases incorporating regulatory changes, NEP 2020 mandate alignments, and usability improvements
- Backward-compatible API updates with minimum 90-day deprecation notice for any breaking changes

Infrastructure Maintenance (CMC):

- Routine database optimization, index management, query performance tuning
- Storage capacity monitoring and proactive scaling triggers
- Backup verification and quarterly restore drills
- SSL certificate renewal management

Monitoring & Security:

- 24×7 automated infrastructure monitoring with alerting for uptime deviation, performance degradation, and security anomalies
- Monthly security scan reports shared with state IT team
- CERT-In aligned incident response — any security incident reported to the state within 6 hours of detection

Governance:

- Monthly service review reports: uptime actuals, incident summary, ticket resolution statistics
- Quarterly business review (QBR) with DoE representatives
- Dedicated Account Manager as single point of escalation for all state-level coordination

10.3 Issue Classification & Resolution SLA

Priority	Definition	Response Time	Resolution Time
P1 — Critical	Complete platform outage; total loss of service affecting all users	< 30 Minutes	< 4 Hours
P2 — High	Core feature broken (login, course access, exam submission, live sessions)	< 2 Hours	< 8 Hours
P3 — Medium	Non-critical feature degraded; workaround available	< 4 Hours	< 24 Hours
P4 — Low	Minor UI issues; configuration assistance; enhancement requests	< 8 Business Hours	Next Sprint Cycle

11. SCALABILITY & FUTURE UPGRADES

11.1 Horizontal Scalability Architecture

The Shiksha Setu LMS is designed from its infrastructure foundations to scale horizontally without architectural changes. Kubernetes-managed microservices allow individual application components — course content delivery, live session management, analytics processing, and notification dispatch — to be scaled independently based on real-time load patterns.

This means the platform can absorb sudden spikes in concurrent users (such as during board exam result announcements, scholarship application windows, or state-wide assignment deadlines) without service degradation or manual intervention.

11.2 Benchmarked Capacity

Metric	Base Configuration	Scaled Configuration
Concurrent Users	Up to 50,000	500,000+
Registered Institutions	Unlimited (multi-tenant)	Unlimited
Content Storage	Elastic (S3-backed)	No hard ceiling
Concurrent Video Streams	Up to 10,000	Unlimited (with CDN scaling)
API Requests per Minute	50,000+	100,000+ (with rate-limit tuning)
Database Query Response (p95)	< 200 milliseconds	< 200 milliseconds

All scaling operations are handled transparently by the platform operations team. No state IT team intervention is required for routine scaling events.

11.3 Roadmap — Planned Enhancements

The following capabilities are on the Shiksha Setu LMS product roadmap for upcoming release cycles:

AI-Powered Learning Recommendations: A personalization engine that analyzes individual student performance trajectories, engagement patterns, and assessment outcomes to surface contextually relevant supplementary content, practice exercises, and remedial modules — without instructor intervention.

Advanced Online Proctoring: AI-assisted exam proctoring with facial presence detection, browser activity monitoring, and suspicious behavior flagging. Designed for high-stakes assessments and board-level examinations conducted through the platform.

Offline Learning Mode (PWA): Progressive Web App capabilities allowing students in low-connectivity or no-connectivity environments to download course content for offline consumption, with automatic synchronization of progress and submissions when connectivity is restored. Critical for equitable access in rural geographies.

Parent / Guardian Portal: A dedicated interface for guardians to monitor student attendance, assignment completion, assessment performance, and receive automated alerts for academic events — without exposing internal platform controls.

National Academic Bank of Credits (ABC) Integration: Compliance with NEP 2020's credit transfer framework through direct integration with the National Academic Depository, enabling students to accumulate, transfer, and redeem credits across institutions seamlessly through the LMS.

Multilingual AI Content Localization: An automated content translation and regional language adaptation engine to convert uploaded course materials into state-specified languages, reducing the content localization burden on teachers in multilingual state deployments.

11.4 Available Add-On Modules (Immediate Integration)

Module	Description	Integration Complexity
3D Scholar	AR/VR immersive STEM content — 10,000+ 3D models, 2,000+ simulations	Low (SSO + API)
AI Teacher Evaluation	Automated grading for handwritten and digital answer sheets	Low (SSO + API)
Virtual Labs	Cloud-based interactive science experiment simulations	Low (SSO + API)
ERP for Schools	End-to-end administration: fees, payroll, timetable, KPI tracking	Medium (data sync)
Digital Library System	Physical and digital library catalog with RFID/barcode support	Low (SSO + API)

All add-on modules share the same SSO authentication layer, meaning students and teachers access the full Shiksha Setu ecosystem through a single login — no separate credentials required.

12. COMPLIANCE & SECURITY STANDARDS

12.1 Security Architecture Philosophy

Shiksha Setu implements a defense-in-depth security model across four layers: identity and access, data protection, network security, and operational governance. No single point of compromise can expose the full system. All security measures are aligned with India's Information Technology Act 2000, CERT-In guidelines, and GDPR-equivalent data protection principles applicable to educational institutions handling student personal data.

12.2 Identity & Access Management

Control	Implementation	Compliance Standard
User Authentication	OTP-based login via Devise gem; passwordless by design	Eliminates credential theft vectors
API / GraphQL Authentication	JWT (JSON Web Tokens) with signed, short-expiry tokens	RFC 7519 compliant
Role-Based Access Control	Granular RBAC — separate permission matrices per role per module	Principle of Least Privilege
Token Lifecycle Management	Token expiration + refresh token rotation; invalidation on logout	OWASP Session Management
Admin Panel Access	IP whitelisting + RBAC + MFA (configurable)	Additional perimeter control
Audit Logging	All authentication events, role changes, and admin actions logged	IT Act 2000 / CERT-In

12.3 Data Encryption Standards

Data in Transit: TLS 1.2 minimum (TLS 1.3 preferred) enforced across all client-server communication, API calls, and inter-service communication via Cloudflare SSL termination and internal mTLS where applicable.

Data at Rest: AES-256 encryption applied at the application layer for all sensitive data fields within PostgreSQL. Database-level encryption is supplemented by application-level field encryption for personally identifiable information.

Object Storage Security: All files stored in Amazon S3 are encrypted using AES-256. Encryption key management is handled centrally by the Rails application, meaning raw encrypted objects in S3 are inaccessible without application-layer key decryption — even to infrastructure-level access.

Key Management: Encryption keys are stored separately from encrypted data, rotated on a defined schedule, and access-restricted to the minimum required application processes.

12.4 Regulatory Compliance Mapping

Regulation / Standard	Compliance Approach
Information Technology Act, 2000 (India)	Data localization options available; audit trail maintained; incident reporting protocol aligned with CERT-In mandates
CERT-In Cybersecurity Guidelines	Vulnerability assessment and penetration testing on schedule; security incident reporting within 6 hours
NEP 2020	Curriculum framework alignment; credit system readiness; multilingual interface support; ABC integration roadmap
Personal Data Protection (PDPA-aligned)	Data minimization practices; consent-based collection; right to erasure workflow; 30-day backup retention with secure disposal
GDPR-Style Data Governance	Privacy-by-design architecture; data processing agreements available for institutional partnerships
Multi-Tenant Data Isolation	Architectural guarantee of no cross-tenant data leakage; separate logical namespaces per institution
RTE (Right to Education) Compliance	Reporting capabilities aligned with RTE attendance and learning outcome monitoring requirements

12.5 Network & Infrastructure Security

Web Application Firewall (WAF): Cloudflare WAF deployed at the edge layer provides protection against OWASP Top 10 vulnerabilities, SQL injection, cross-site scripting (XSS), and bot-driven attacks.

DDoS Mitigation: Cloudflare's anycast network absorbs volumetric DDoS attacks at the edge, preventing them from reaching application infrastructure. No additional DDoS appliance required.

Rate Limiting: API gateway enforces per-IP, per-token, and per-endpoint rate limiting to prevent brute force attacks, credential stuffing, and abusive scraping.

Container Security: Kubernetes pod security policies restrict container privilege levels. No container runs with root privileges in production. Image scanning integrated into CI/CD pipeline via GitLab's security scanning tools.

Dependency Management: Automated Gemfile and package.json vulnerability scanning on every commit. Critical vulnerabilities trigger immediate pipeline failures and engineering alerts.

12.6 Vulnerability Management & Audit

- Automated SAST (Static Application Security Testing) and DAST (Dynamic Application Security Testing) integrated into GitLab CI/CD pipeline — every code commit is security-scanned before deployment
- Periodic third-party penetration testing by CERT-In empaneled security firms; reports available to DoE upon request
- Responsible vulnerability disclosure policy published; defined remediation SLAs
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13. WARRANTY & SLA DETAILS

13.1 Software Warranty

Warranty Item	Terms
Warranty Period	12 months from the date of production go-live
Software Defects	All reported bugs and functional defects corrected at no additional charge during warranty period
Security Vulnerabilities	Critical vulnerabilities (CVSS 9.0+) patched within 24 hours; High severity within 72 hours; Medium within 14 days
Data Integrity	Any data loss attributable to platform failure remediated at no cost; full restoration from the nearest successful backup
Performance Degradation	If platform consistently fails to meet agreed performance benchmarks, remediation is prioritized at P2 level regardless of visible user impact
Post-Warranty Continuity	All warranty terms continue uninterrupted under an active AMC/CMC agreement

13.2 Service Level Agreement (SLA)

SLA Metric	Committed Level
Platform Uptime	99.5% monthly availability per module (equivalent to less than 3.6 hours downtime per month)
Scheduled Maintenance Window	Sundays 2:00 AM – 5:00 AM IST; minimum 48-hour advance notice to DoE
Emergency Maintenance	Immediate notification via State Nodal Officer Hotline; maximum 4-hour window
Data Backup Frequency	Daily automated backups; 30-day retention
Recovery Point Objective (RPO)	24 hours for standard data; 4 hours for transaction-critical data
Disaster Recovery — RTO	< 4 hours for full platform restoration from backup in a declared DR scenario
Core API Availability	99.9% for critical endpoints (authentication, course access, assessment submission)
Page Load Performance	< 3 seconds on standard broadband (10 Mbps+); < 6 seconds on 4G mobile
Content Delivery (Cloudflare CDN)	Global cache hit ratio > 85%; media assets served from nearest edge node

13.3 SLA Breach Remedies

In the event of SLA breaches, the following remedies apply and are contractually enforced:

Breach Scenario	Remedy	Measurement Period
Monthly uptime < 99.5%	5% credit on that month's applicable fee	Monthly
Monthly uptime < 99.0%	10% credit on that month's applicable fee	Monthly
Monthly uptime < 98.0%	15% credit + right to terminate without penalty	Monthly
P1 response time exceeded	Per-incident service credit	Per incident
Data loss due to platform failure	Full data restoration + 15% credit on monthly fee	Per incident
RTO not met in DR scenario	20% credit on monthly fee for affected period	Per incident

Service credits are applied to the following billing cycle and do not constitute the sole remedy — the Department retains the right to escalate unresolved SLA failures as contractual breaches.

13.4 Business Continuity & Disaster Recovery

Primary — Secondary Failover: Application services are deployed across multiple availability zones. A primary availability zone failure triggers automatic failover to a secondary zone within 15 minutes without manual intervention.

Backup Architecture: Daily encrypted snapshots of PostgreSQL databases and all S3 object storage. Backups are stored in a geographically separate region from the primary deployment. Backup integrity is verified weekly through automated restore tests.

Incident Communication: A dedicated status page is maintained and updated in real time during any P1 or P2 incident. All registered state and district administrators are automatically notified via email and SMS when the status page is updated.

14. COMMERCIAL BREAKUP

The commercial details for this engagement — including implementation costs, annual license fees, AMC/CMC rates, and optional module pricing — will be shared separately through a formal Commercial Proposal following a detailed scoping discussion with the Department.

All indicative commercial information to be provided in the formal proposal is subject to:

- Final user volume and institution count confirmation
- Selected deployment model (SaaS / Private Cloud / Hybrid)
- Integration complexity assessment
- Contract duration and payment structure preferences
- Applicable government procurement norms (GeM listing / QCBS / L1 tender requirements)

[TBD — Formal Commercial Proposal to be Shared Separately]

15. CONCLUSION & NEXT STEPS

15.1 Summary

The Shiksha Setu Learning Management System represents a comprehensive, enterprise-grade digital education infrastructure purpose-designed for state-wide government deployment. It brings together course delivery, user management, assessments, analytics, communications, financial management, and career support into a single, unified, secure, and scalable platform — while remaining modular enough to accommodate the diverse needs of different institution types, levels, and geographies within the state.

The platform's architecture is built for the long term: cloud-native for scalability, API-first for integration flexibility, and security-by-design for the rigorous compliance demands of government education data governance.

Beyond technology, Shiksha Setu's implementation methodology, training programs, and support model are structured to ensure that the platform genuinely transforms how teachers teach and students learn — not merely providing tools that remain underutilized due to inadequate adoption support.

15.2 Why Shiksha Setu LMS

Dimension	Shiksha Setu Advantage
Regulatory Alignment	Built with NEP 2020, IT Act 2000, CERT-In, and RTE compliance as design principles — not afterthoughts
Government-Grade Security	Defense-in-depth security model; AES-256 + TLS 1.2+ + JWT + RBAC + WAF — all standard
True Multi-Tenancy	Architectural isolation per institution — not just application-level access controls
Integration Readiness	REST + GraphQL APIs, bulk import, SSO/OAuth — connects with any state MIS or ERP system
Scalability Without Ceiling	Kubernetes-native horizontal scaling; tested to 500,000+ concurrent users
End-to-End Ecosystem	Optional modular add-ons (3D Scholar, AI Evaluation, Virtual Labs, ERP, Library) expand capability without platform change
Sustainable Adoption	Train-the-Trainer model, multilingual support, and embedded digital training ensure genuine state-wide adoption

15.3 Proposed Next Steps

Step 1 — Live Product Demonstration

A comprehensive demonstration of the full LMS platform — including student, teacher, admin, and state-level dashboards — conducted for the Department's technical and academic evaluation committee.

Step 2 — Scoping Workshop

A structured half-day workshop with DoE representatives and district coordinators to finalize: institution count, user volumes, integration requirements, deployment preferences, and academic calendar alignment.

Step 3 — Formal Commercial Proposal

Following the scoping workshop, a detailed Commercial Proposal — including final pricing, contract structure, milestone-based payment schedule, and SLA terms — will be submitted to the Department within 7 business days.

Step 4 — Pilot Agreement

A Pilot Agreement for 5–10 seed institutions can be executed ahead of the full state contract, allowing the Department to evaluate the platform in a live environment with real users before committing to state-wide procurement.